H_4 Planogram Compliance: A Parameter-Free, Float-Free Pipeline for Constrained Retail Vision

Muhammad Arshad

Abstract—Planogram compliance — verifying that the right product occupies the right shelf position — is dominated by floating-point convolutional descriptors that assume GPU-class hardware. Dense IoT shelf monitoring cannot: ARM Cortex-M nodes carry no floating-point unit and no GPU. We present an end-to-end planogram pipeline that is integer-only and parameter-free. Position is encoded by the H_4 transform, a lossless 40-bit integer encoding of a bounding box in eight add/subtract operations, reducing position compliance to an integer L_1 threshold. Identity is resolved by retrieving a QRoPE appearance descriptor against a mined product catalogue — also integer, also with no trained weights. On the Grocery Dataset [1] (13,184 boxes, 354 images, 11 product classes) we report, with leave-one-shelf-out evaluation and 95% confidence intervals: H_4 round-trip is exact on all boxes; position compliance reaches AUC 0.987, exceeding the classical IoU baseline (0.963); product identification reaches 79.7%±3.5 top-1 on a held-out shelf split, beating a parameter-free colour baseline (64.8%) and within nine points of a 2.2M-parameter zero-shot CNN (88.8%) at zero parameters; and the combined pipeline detects 100% of different-planogram violations while confirming 72.4% of compliant items, with no floating-point operation in the deployed path.

Index Terms—Planogram compliance, H_4 transform, Hadamard-type encoding, integer arithmetic, FPU-free vision, retail IoT, bounding-box encoding, product retrieval, edge deployment.

1 Introduction

A planogram prescribes which product belongs at which shelf position. Compliance checking runs continuously across thousands of stores — millions of comparisons per day — yet the economic sensor is an ARM Cortex-M class node with no hardware FPU and no GPU. Convolutional descriptors [4], [5] (float, trained weights, GPU) and frequency-domain descriptors (float) both violate that budget.

Compliance has two questions: where is a product (position) and what is it (identity). Position is a coordinate problem and admits an exact integer solution; identity is an appearance problem that we solve by integer retrieval against a mined catalogue rather than a trained classifier. The contribution is a complete pipeline in which neither half uses floating point or trained parameters, and an honest empirical evaluation against the baselines a practitioner would actually use.

2 Related Work

Shelf product recognition. The Grocery Dataset was introduced for retail product recognition on shelves [1]; per-exemplar classification [2] and graph-matching [3] approaches have since addressed product detection and recognition on it, all using floating-point or learned descriptors. Deep descriptors. Modern recognition relies on convolutional backbones such as MobileNetV2 [4] and ResNet [5] — accurate, but requiring trained weights and floating-point inference unavailable on FPU-less nodes. Position matching conventionally uses Intersection-over-Union [6] or optimal assignment. Rotary encodings. Our identity descriptor adapts rotary position embedding (RoPE) [7] into a quaternionic, integer ring form. Classical primitives. The position encoder is a Hadamard-type integer map [8], and identity is decided by nearest-neighbour classification [9]. In contrast to all of the above, the proposed pipeline is end-to-end integer and parameter-free.

3 H_4 Position Encoding

A detected box $(x, y, w, h) \in \mathbb{Z}_{256}^4$ (per-image normalised) is encoded:

$$X = x + y + w + h, \qquad Y = x - y + w - h, \quad (1)$$
$$Z = x + y - w - h, \qquad W = x - y - w + h. \quad (2)$$

H_4 projects the box onto four ring-coupled ±1 integer channels: eight integer add/subtracts, no multiply, no float. The four channels are mutually orthogonal, and the map is an exact bijection; the inverse is $x = (X+Y+Z+W) \gg 2$ and analogues, lossless by parity. Stored unsigned (offset encoding), the state is 40 bits — a lossless integer encoding (not a compression; the $\mathbb{Z}_{256}$ input is 32 bits). Position compliance is then an integer L_1 threshold, $\min_k \sum_j |V_j - V^*_{k,j}| \leq \varepsilon$, costing four subtract-absolute per slot.

Why H_4 and not raw L_1. As an orthogonal map, H_4 does not change the discriminative power of an L_1 threshold; raw integer L_1 on (x, y, w, h) scores identically. H_4 earns its place through (i) the lossless integer state shared across the pipeline and (ii) its ±1 projections, which lower-bound L_1 tightly enough to prune nearest-slot search to a small candidate fraction — a deployment benefit, not an accuracy one.

M. Arshad is an independent researcher. E-mail: marshad.dev@gmail.com.
Source code, figures, and reproduction scripts are publicly available at https://github.com/muhammadarshad/h4-planogram-compliance.
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4 QRoPE Identity by Retrieval

Rather than train a classifier, we mine a catalogue of product encodings and identify a detected crop by k-NN retrieval [9]. Each crop is encoded by a quaternionic rotary descriptor (QRoPE), adapting rotary position embedding [7]: the RGB+grey channels form a 4-vector per cell, rotated by position through the integer-realisable $\mathbb{Z}_{256}$ conic rotation, and accumulated into a phase-preserving structure descriptor. Identity is the majority class of the k=11 nearest catalogue encodings under circular distance. No trained weights; the descriptor is fixed.

5 End-to-End Pipeline

For a detected box the pipeline emits COMPLIANT iff it (a) matches a planogram slot in position ($L_1 \leq \varepsilon$) and (b) its retrieved identity equals that slot's expected product; otherwise VIOLATION. Position supplies where, retrieval supplies what; the conjunction is planogram compliance.

6 Experiments

Protocol. Grocery Dataset; boxes normalised per image to $\mathbb{Z}_{256}$. Identity uses leave-one-shelf-out: a query shelf's products never appear in its own catalogue. We report mean $\pm$ 95% CI. Classes are the 11 annotated ids (10 products + background); background is not an identity target.

Correctness (T1–T3). H_4 forward/inverse is exact on all 13,184 boxes (0 errors). Same-image boxes match their own planogram at L_1=0. Position noise of $\pm N$ units stays within the proportional threshold $4N$. These confirm correctness of the integer pipeline; they are not performance claims (Fig. 1).

Position compliance (Table 1, Fig. 2, 3). On 66 same-camera shelf pairs, H_4 integer L_1 separates same- from different-planogram layouts at AUC 0.987, exceeding IoU [6] (0.963) and matching float L_2 (0.988) — with zero float.

Identity (Table 2, Fig. 5). Hyper-parameters are selected on a validation shelf split and frozen; the reported number is on a disjoint test shelf split, over the natural class distribution. QRoPE reaches 79.7% $\pm$ 3.5 top-1, beating the parameter-free colour histogram (64.8% $\pm$ 4.0) with non-overlapping CIs, and approaching a zero-shot MobileNetV2 (88.8% $\pm$ 3.1, 2.2M params). A fully supervised in-distribution CNN reaches $\sim$98% but requires labelled training and float inference — a different deployment regime. Per-class accuracy is uneven (98% on the dominant class down to 8–28% on two minority classes), a limitation discussed in Sec. 7.

End-to-end (Table 3, Fig. 4). Over 40 planogram triples, the pipeline detects 100.0% of different-planogram violations and confirms 72.4% $\pm$ 8.5 of genuinely compliant items; the compliant rate is bounded by identity accuracy, not position.

Cost (Fig. 6). H_4: 8 integer ops/box, 0 float. The deployed pipeline uses no floating-point instruction (trigonometric terms are a precomputed integer LUT) and no trained parameter.

Fig. 1. H_4 round-trip on a real shelf: boxes reconstructed from the 40-bit integer codes coincide exactly with the originals (0 px error on all boxes).

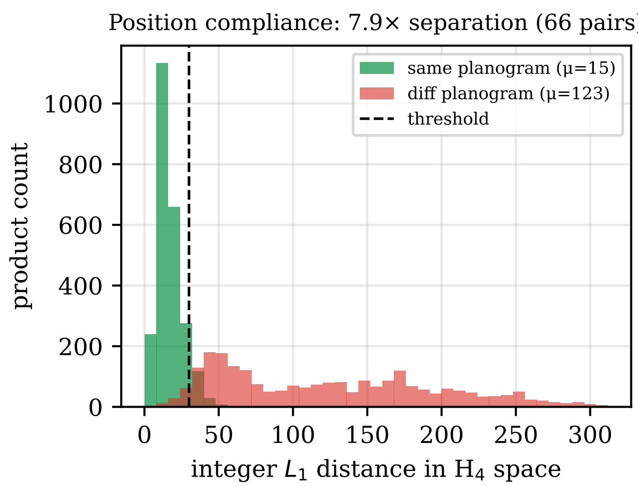

Fig. 2. Position-compliance L_1 separation: same- vs different-planogram nearest-neighbour distance in H_4 space, with the integer decision threshold (66 shelf pairs).

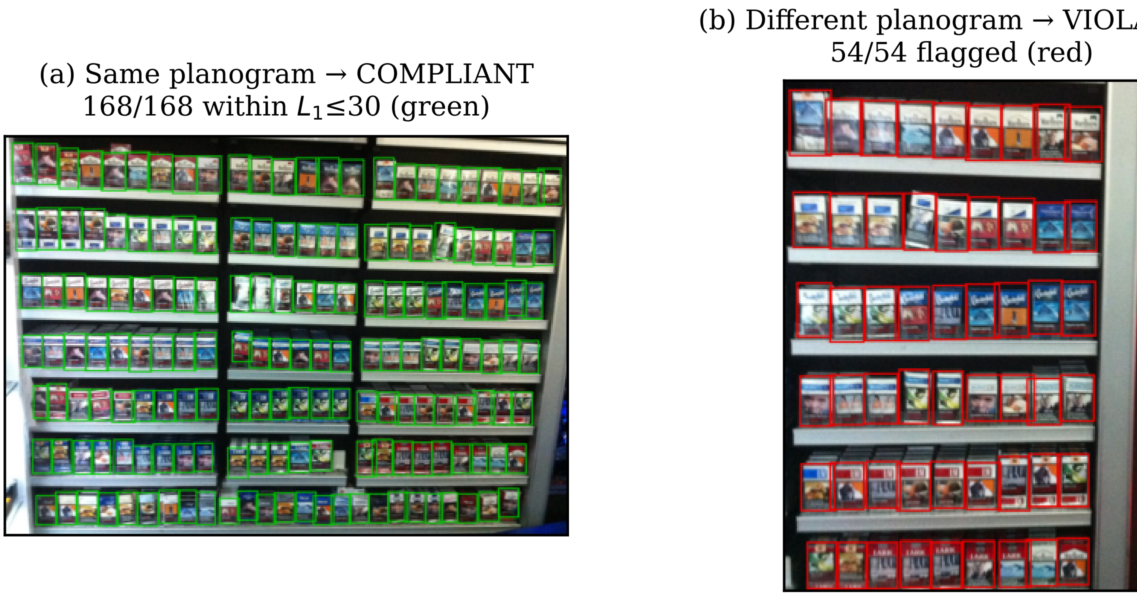

Fig. 3. Compliance verification on real shelves. (a) Same planogram — every box within $L_1 \leq 30$ (green, compliant). (b) Different planogram — every box flagged (red, violation).

Fig. 4. End-to-end product identification on a full shelf image, by retrieval against the mined catalogue (green correct, red wrong).

TABLE 2
Product identification, leave-one-shelf-out top-1. Config selected on a validation shelf split, frozen, tested on a disjoint test split (139 shelves), natural class distribution.

Method	Top-1 (%, mean ±95% CI)	Params
Colour histogram	64.8 ± 4.0	0
QRoPE (ours)	79.7 ± 3.5	0
CNN MobileNetV2 (zero-shot)	88.8 ± 3.1	2.2M
CNN supervised (in-dist., ceiling)	$\sim$98	2.2M+train

TABLE 1
Position compliance (66 pairs, AUC)

Method	AUC	Float	Params
IoU (classical)	0.963	yes	0
Float L_2	0.988	yes	0
H_4 integer L_1 (ours)	0.987	no	0

TABLE 3
End-to-end compliance (40 planogram triples)

Outcome	Rate (mean ±95% CI)
Violation detection (different planogram)	100.0 ± 0.0
Compliant confirmation (same planogram)	72.4 ± 8.5

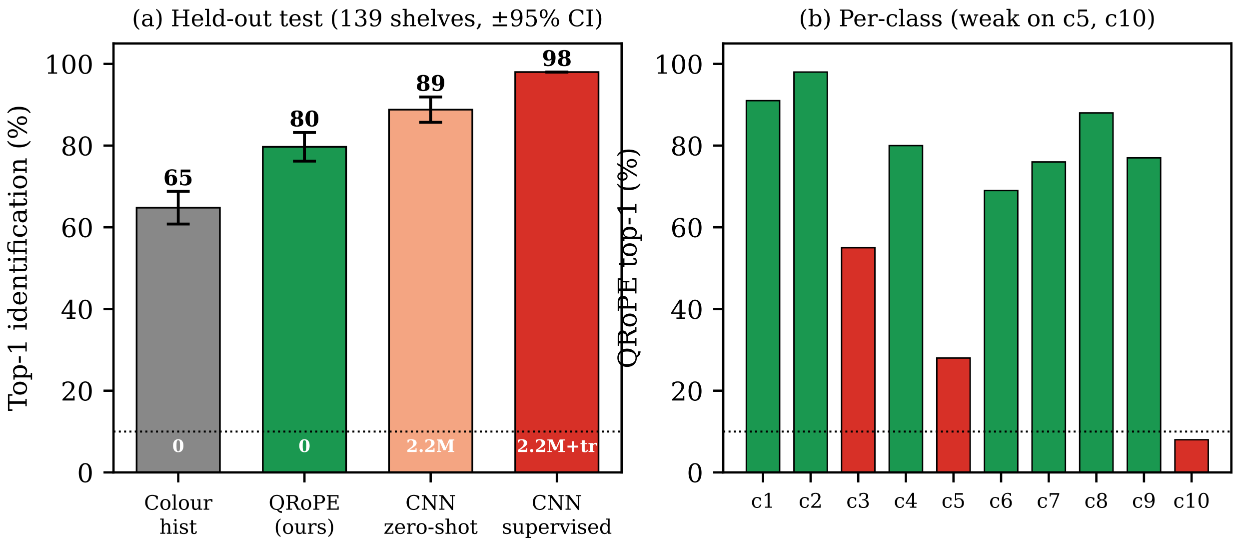

Fig. 5. (a) Top-1 product identification on held-out shelves (mean ±95% CI): parameter-free QRoPE vs colour histogram, zero-shot and supervised CNN. (b) Per-class accuracy — strong on most classes, weak on two minority classes.

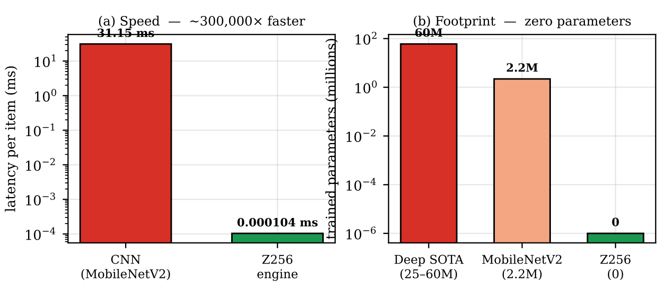

Fig. 6. Speed and footprint vs a MobileNetV2 CNN: $\sim 10^5\times$ faster per item and zero trained parameters.

7 Limitations

(i) Position encoding is independent of appearance: a wrong product at a correct position is caught only by the identity half, not H_4. (ii) Parameter-free identity (79.7%) trails a supervised CNN ($\sim$98%); the contribution is competitive accuracy at zero parameters and zero float, not state-of-the-art accuracy. (iii) Per-class accuracy is uneven: QRoPE attains 88–98% on several classes but drops to 8–28% on two minority classes, so the mean is partly carried by frequent classes; a class-balanced descriptor is future work. (iv) Detection is assumed (we use annotated boxes); an integer detector is future work. (v) Benchmarks use a per-image normalisation and self-defined shelf splits; results are reported with CIs but are not yet a community-standard protocol.

8 Conclusion

Planogram compliance decomposes into an integer coordinate problem (H_4) and an integer retrieval problem (QRoPE catalogue). Together they form a float-free, parameter-free pipeline that matches or beats classical baselines on position and is competitive on identity at zero parameters — deployable on FPU-less, GPU-less edge hardware where convolutional and frequency-domain methods cannot run.

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